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Deep Generative Models

Lecture 5

Roman Isachenko

Moscow Institute of Physics and Technology
Yandex School of Data Analysis

Recap of Previous Lecture

Theorem

$$\frac{1}{n}\sum_{i=1}^{n} D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}_i) \,\|\, p(\mathbf{z})) = D_{\mathrm{KL}}(q_{\mathrm{agg},\boldsymbol{\phi}}(\mathbf{z}) \| p(\mathbf{z})) + \mathbb{I}_q[\mathbf{x}, \mathbf{z}].$$

Revisiting the ELBO

$$\frac{1}{n}\sum_{i=1}^{n} \mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}_i) = \underbrace{\frac{1}{n}\sum_{i=1}^{n} \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}_i)} \log p_{\boldsymbol{\theta}}(\mathbf{x}_i|\mathbf{z})}_{\text{Reconstruction Loss}} - \underbrace{\mathbb{I}_q[\mathbf{x}, \mathbf{z}]}_{\text{Mutual Information}} - \underbrace{D_{\mathrm{KL}}(q_{\mathrm{agg},\boldsymbol{\phi}}(\mathbf{z}) \| p(\mathbf{z}))}_{\text{Marginal KL}}.$$

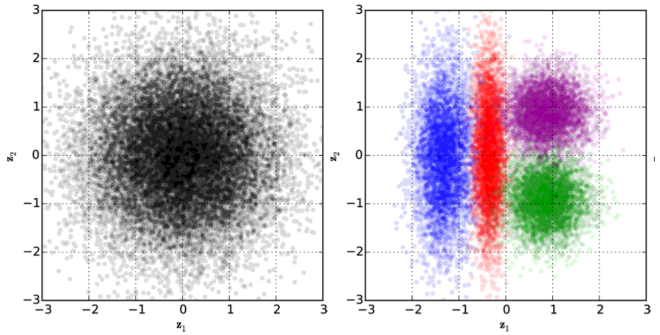
Optimal VAE Prior

$$D_{\mathrm{KL}}(q_{\mathrm{agg},\boldsymbol{\phi}}(\mathbf{z}) \,\|\, p(\mathbf{z})) = 0 \;\Leftrightarrow\; p(\mathbf{z}) = q_{\mathrm{agg},\boldsymbol{\phi}}(\mathbf{z}) = \frac{1}{n}\sum_{i=1}^{n} q_{\boldsymbol{\phi}}(\mathbf{z}|\mathbf{x}_i).$$

Thus, the optimal prior distribution $p(\mathbf{z})$ is the aggregated variational posterior $q_{\mathrm{agg},\boldsymbol{\phi}}(\mathbf{z})$.

Recap of Previous Lecture

- **Prior mismatch:** The unimodal encoder $q_{\phi}(\mathbf{z}|\mathbf{x})$ yields $q_{\text{agg},\phi}(\mathbf{z})$ that often does not match the Gaussian prior $p(\mathbf{z})$.
- **Blurriness from averaging:** With a fixed encoder, prior and Gaussian decoder variance, ELBO maximization gives $\boldsymbol{\mu}^*(\mathbf{z}) = \mathbb{E}_{q_{\phi}(\mathbf{x}|\mathbf{z})}[\mathbf{x}]$. If distinct inputs $\mathbf{x} \neq \mathbf{x}'$ map to *overlapping* latent regions, the decoder averages over unrelated data.
- **Fixed encoder and prior:** The conditional KL is constant in $\boldsymbol{\theta}$, not necessarily zero. At $p = q_{\text{agg},\phi}$, only the marginal KL vanishes; the average conditional KL equals $\mathbb{I}_q[\mathbf{x}, \mathbf{z}]$.



(a) Prior distribution (b) Posteriors in standard VAE

Rezende D. J., Mohamed S. Variational Inference with Normalizing Flows, 2015
Lai C. H. et al. The principles of diffusion models, 2025.

Recap of Previous Lecture

Assumptions

- Let $c \sim \mathrm{Cat}(\boldsymbol{\pi})$, where $\pi_k = P(c = k)$ and $\sum_{k=1}^{K} \pi_k = 1$.
- For tokenizer training, fix $p(c) = \mathrm{Uniform}\{1, \dots, K\}$.

ELBO

$$\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) = \mathbb{E}_{q_{\boldsymbol{\phi}}(c|\mathbf{x})} \log p_{\boldsymbol{\theta}}(\mathbf{x}|c) - D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(c|\mathbf{x}) \,\|\, p(c)) \to \max_{\boldsymbol{\phi},\boldsymbol{\theta}}.$$

$$D_{\mathrm{KL}}(q_{\boldsymbol{\phi}}(c|\mathbf{x}) \,\|\, p(c)) = -\mathrm{H}(q_{\boldsymbol{\phi}}(c|\mathbf{x})) + \log K.$$

Quantized Representation

Define the codebook (dictionary) space $\{\mathbf{e}_k\}_{k=1}^{K}$ with $\mathbf{e}_k \in \mathbb{R}^L$ and K the number of codebook entries.

$$\mathbf{z}_q = \mathbf{q}(\mathbf{z}) = \mathbf{e}_{k^*}, \quad \text{where } k^* = \arg\min_k \|\mathbf{z} - \mathbf{e}_k\|.$$

Codebook collapse: Only a small subset of entries is used across the data; restart an unused code at an encoder output.

Oord A., Vinyals O., Kavukcuoglu K. Neural Discrete Representation Learning, 2017
Zhao Y. et al. Feature Quantization Improves GAN Training, 2020 · Mentzer F. et al. Finite Scalar Quantization, 2023

Recap of Previous Lecture

Quantize each of the $M = WH$ encoder vectors using the **same codebook**; the code map is $\mathbf{c} = (c_1, \dots, c_M)$, with $\mathbf{z}_{q,m} = \mathbf{e}_{c_m}$.

Deterministic Variational Posterior

$$q_{\boldsymbol{\phi}}(c_m = k|\mathbf{x}) = \begin{cases} 1, & \text{for } k = \arg\min_j \|\mathbf{z}_{e,m} - \mathbf{e}_j\|; \\ 0, & \text{otherwise.} \end{cases}$$

ELBO

$$\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) = \mathbb{E}_{q_{\boldsymbol{\phi}}(\mathbf{c}|\mathbf{x})} \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}_q(\mathbf{c})) - M \log K = \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}_q) - M \log K.$$

Straight-Through Gradient Estimator

$$\frac{\partial \log p(\mathbf{x}|\mathbf{z}_q, \boldsymbol{\theta})}{\partial \boldsymbol{\phi}} = \frac{\partial \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}_q)}{\partial \mathbf{z}_q} \cdot \frac{\partial \mathbf{z}_q}{\partial \boldsymbol{\phi}} \approx \frac{\partial \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}_q)}{\partial \mathbf{z}_q} \cdot \frac{\partial \mathbf{z}_e}{\partial \boldsymbol{\phi}}$$

The independent uniform prior is used during tokenizer training. Then fix the tokenizer and decoder, and fit an autoregressive prior $p_{\boldsymbol{\psi}}(\mathbf{c})$ to the aggregated posterior $q_{\text{agg},\boldsymbol{\phi}}(\mathbf{c})$.

Recap of Previous Lecture

Training

Stage 1: tokenizer.

1. Sample $\mathbf{x} \sim p_{\text{data}}(\mathbf{x})$.
2. Compute encoder outputs $\mathbf{z}_{e,m} = \text{NN}_{e,\boldsymbol{\phi}}(\mathbf{x})_m$.
3. Compute the quantized representation at each location:
$$c_m = \arg\min_k \|\mathbf{z}_{e,m} - \mathbf{e}_k\|_2, \quad \mathbf{z}_{q,m} = \mathbf{e}_{c_m}.$$
4. Compute the ELBO with the uniform training prior:
$$\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) = \log p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}_q) - M \log K.$$
5. Compute the loss with codebook and commitment terms:
$$\mathcal{L} = -\mathcal{L}_{\boldsymbol{\phi},\boldsymbol{\theta}}(\mathbf{x}) + \sum_{m=1}^{M} \left\|\text{sg}[\mathbf{z}_{e,m}] - \mathbf{e}_{c_m}\right\|_2^2 + \beta \sum_{m=1}^{M} \left\|\mathbf{z}_{e,m} - \text{sg}[\mathbf{e}_{c_m}]\right\|_2^2.$$
6. Update $\boldsymbol{\phi}$, $\boldsymbol{\theta}$ and $\{\mathbf{e}_k\}$ (straight-through for $\boldsymbol{\phi}$).

Sampling

After **Stage 2: prior training**, fix all parameters.

1. Sample the code map $\mathbf{c} \sim p_{\boldsymbol{\psi}}(\mathbf{c})$ autoregressively.
2. Compute $\mathbf{z}_{q,m} = \mathbf{e}_{c_m}$ at each location.
3. Sample $\mathbf{x} \sim p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}_q)$.

Recap of Previous Lecture

Likelihood-Free Learning

- Likelihood isn't always a suitable metric for evaluating generative models.
- Sometimes, the likelihood function can't even be computed exactly.

Imagine we have two sets of samples:

- $\{\mathbf{x}_i\}_{i=1}^{n_1} \sim p_{\text{data}}(\mathbf{x})$ — real samples;
- $\{\mathbf{x}_i\}_{i=1}^{n_2} \sim p_{\boldsymbol{\theta}}(\mathbf{x})$ — generated (fake) samples.

$$p(y = 1|\mathbf{x}) = P(\mathbf{x} \sim p_{\text{data}}(\mathbf{x})); \quad p(y = 0|\mathbf{x}) = P(\mathbf{x} \sim p_{\boldsymbol{\theta}}(\mathbf{x}))$$

Assumption

The generative model $p_{\boldsymbol{\theta}}(\mathbf{x})$ matches $p_{\text{data}}(\mathbf{x})$ if a discriminative model $p(y|\mathbf{x})$ can't distinguish between them — that is, if $p(y = 1|\mathbf{x}) = 0.5$ for every $\mathbf{x}$.

- **Generator:** a generative model $\mathbf{x} = \mathbf{G}_{\boldsymbol{\theta}}(\mathbf{z})$ that produces more realistic samples.
- **Discriminator:** a classifier $D_{\boldsymbol{\phi}}(\mathbf{x}) \in [0, 1]$ distinguishing real from generated samples.

Recap of Previous Lecture

Log-Likelihood for Discriminative Model

$$\max_{p(y|\mathbf{x})} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log p(y=1|\mathbf{x}) + \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} \log p(y=0|\mathbf{x})\right]$$

- **Discriminator:** A classifier $p_{\boldsymbol{\phi}}(y=1|\mathbf{x}) = D_{\boldsymbol{\phi}}(\mathbf{x}) \in [0,1]$, distinguishing real and generated samples. The discriminator aims to **maximize** log-likelihood.
- **Generator:** The generative model $\mathbf{x} = \mathbf{G}_{\boldsymbol{\theta}}(\mathbf{z}),\ \mathbf{z} \sim p(\mathbf{z})$, seeks to fool the discriminator. The generator aims to **minimize** the same log-likelihood objective.

GAN Objective

$$\min_G \max_D \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} \log(1 - D(\mathbf{x}))\right]$$

$$\min_G \max_D \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D(\mathbf{G}(\mathbf{z})))\right]$$

Outline

01 Generative Adversarial Networks (GAN)

02 Wasserstein Distance

03 Wasserstein GAN (WGAN)

04 Evaluation of Likelihood-Free Models

Frechet Inception Distance (FID) · Precision-Recall · CLIP Score · Human Eval

Outline

01 **Generative Adversarial Networks (GAN)**

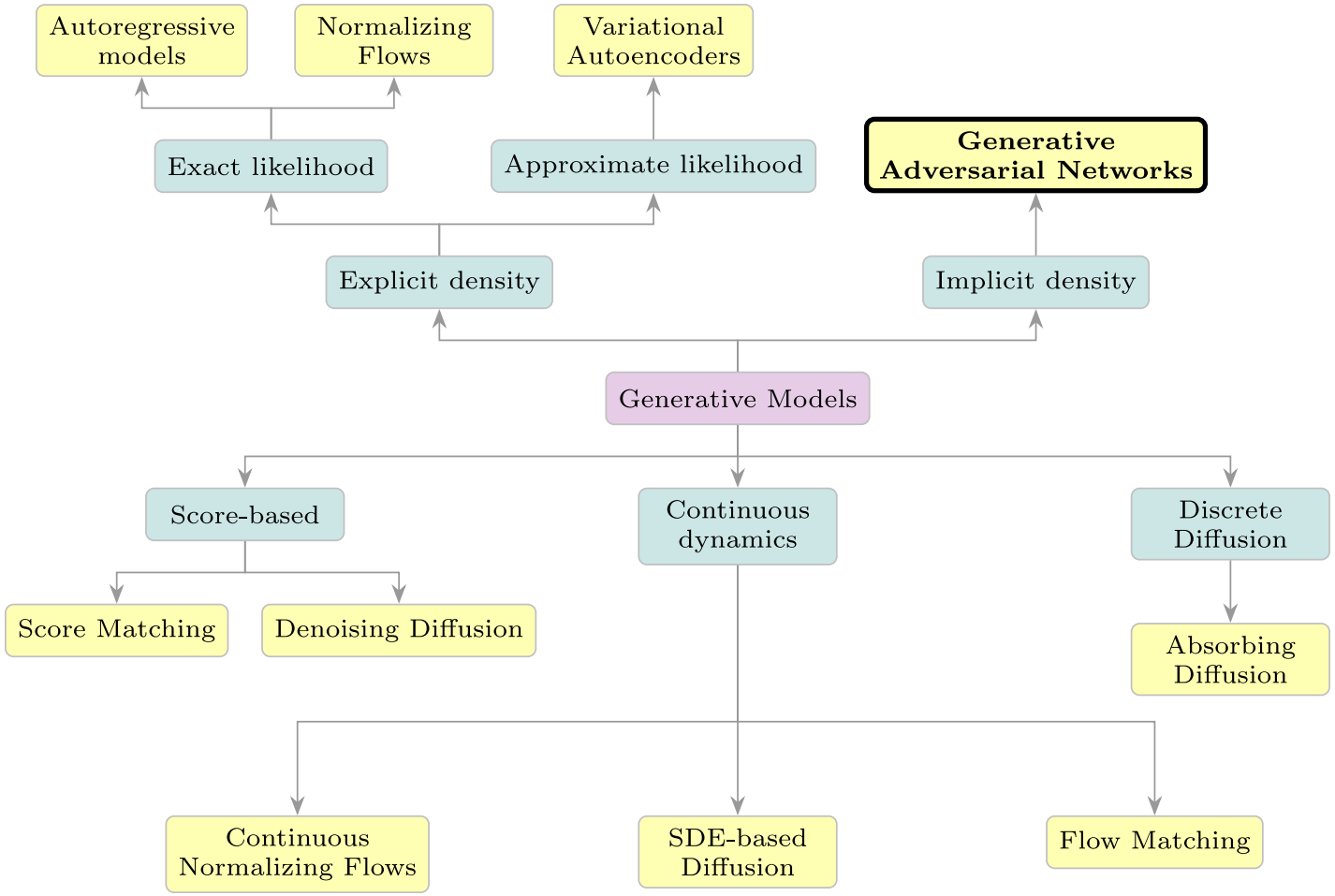
02 Wasserstein Distance

03 Wasserstein GAN (WGAN)

04 Evaluation of Likelihood-Free Models

Frechet Inception Distance (FID) · Precision-Recall · CLIP Score · Human Eval

Generative Models Taxonomy



GAN Optimality

Theorem

The minimax game

$$\min_G \max_D \Big[\underbrace{\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D(\mathbf{G}(\mathbf{z})))}_{V(G,D)}\Big]$$

achieves its global optimum when $p_{\text{data}}(\mathbf{x}) = p_{\boldsymbol{\theta}}(\mathbf{x})$, and $D^*(\mathbf{x}) = 0.5$.

Proof (Fixed G)

$$\begin{aligned} V(G, D) &= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} \log(1 - D(\mathbf{x})) \\ &= \int \underbrace{\left[p_{\text{data}}(\mathbf{x}) \log D(\mathbf{x}) + p_{\boldsymbol{\theta}}(\mathbf{x}) \log(1 - D(\mathbf{x}))\right]}_{y(D)} d\mathbf{x} \end{aligned}$$

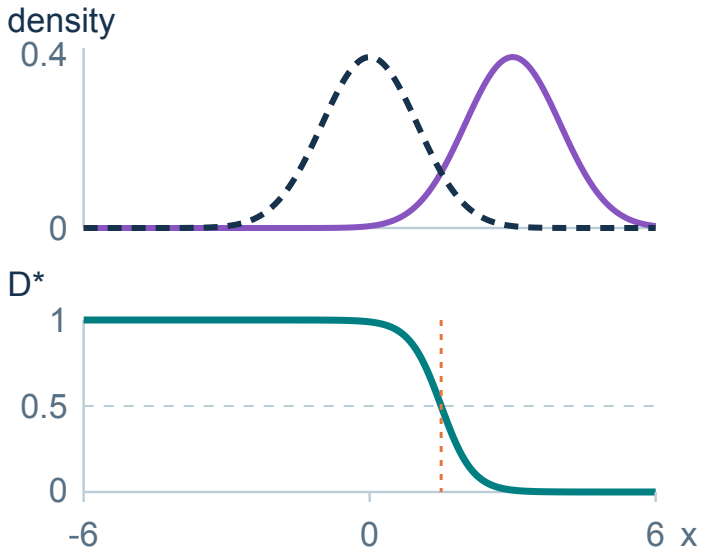
$$\frac{dy(D)}{dD} = \frac{p_{\text{data}}(\mathbf{x})}{D(\mathbf{x})} - \frac{p_{\boldsymbol{\theta}}(\mathbf{x})}{1 - D(\mathbf{x})} = 0 \quad \Rightarrow \quad D^*(\mathbf{x}) = \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_{\boldsymbol{\theta}}(\mathbf{x})}$$

The Optimal Discriminator Responds to the Densities

$$D^*(x) = \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_{\boldsymbol{\theta}}(x)}, \qquad p_{\text{data}}(x) = \mathcal{N}(0, 1), \quad p_{\boldsymbol{\theta}}(x) = \mathcal{N}(\mu, 1).$$

Separated $\mu = 3$

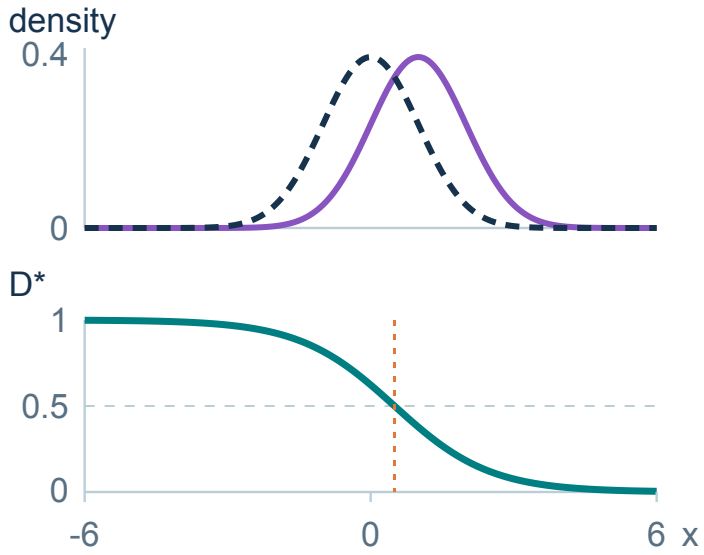
$p_{\text{data}}(x) = \mathcal{N}(0, 1)$ $p_{\boldsymbol{\theta}}(x) = \mathcal{N}(3, 1)$



Equal densities at x = 1.5.

Overlap $\mu = 1$

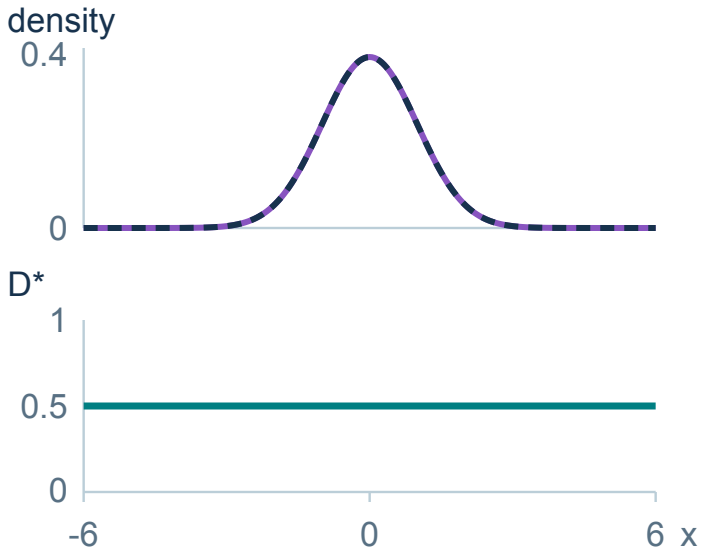
$p_{\text{data}}(x) = \mathcal{N}(0, 1)$ $p_{\boldsymbol{\theta}}(x) = \mathcal{N}(1, 1)$



Equal densities at x = 0.5.

Matched $\mu = 0$

$p_{\text{data}}(x) = \mathcal{N}(0, 1)$ $p_{\boldsymbol{\theta}}(x) = \mathcal{N}(0, 1)$



Equal densities: D* = 0.5 everywhere.

Matching the generator to the data makes the optimal discriminator equal to 0.5 everywhere.

GAN Optimality

Proof Continued (Fixed $D = D^*$)

$$
\begin{aligned}
V(G, D^*) &= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log \left(\frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_{\boldsymbol{\theta}}(\mathbf{x})} \right) + \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} \log \left(\frac{p_{\boldsymbol{\theta}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_{\boldsymbol{\theta}}(\mathbf{x})} \right) \\
&= D_{\text{KL}} \left(p_{\text{data}}(\mathbf{x}) \,\|\, \frac{p_{\text{data}}(\mathbf{x}) + p_{\boldsymbol{\theta}}(\mathbf{x})}{2} \right) + D_{\text{KL}} \left(p_{\boldsymbol{\theta}}(\mathbf{x}) \,\|\, \frac{p_{\text{data}}(\mathbf{x}) + p_{\boldsymbol{\theta}}(\mathbf{x})}{2} \right) - 2 \log 2 \\
&= 2\, D_{\text{JS}}(p_{\text{data}}(\mathbf{x}) \,\|\, p_{\boldsymbol{\theta}}(\mathbf{x})) - 2 \log 2.
\end{aligned}
$$

Jensen-Shannon Divergence (Symmetric KL Divergence)

$$
D_{\text{JS}}(p_{\text{data}}(\mathbf{x}) \| p_{\boldsymbol{\theta}}(\mathbf{x})) = \frac{1}{2} \left[D_{\text{KL}} \left(p_{\text{data}}(\mathbf{x}) \,\|\, \frac{p_{\text{data}}(\mathbf{x}) + p_{\boldsymbol{\theta}}(\mathbf{x})}{2} \right) + D_{\text{KL}} \left(p_{\boldsymbol{\theta}}(\mathbf{x}) \,\|\, \frac{p_{\text{data}}(\mathbf{x}) + p_{\boldsymbol{\theta}}(\mathbf{x})}{2} \right) \right]
$$

Its square root $\sqrt{D_{\text{JS}}(p \| q)}$ is a distance metric.

$$
V(G^*, D^*) = -2 \log 2, \quad p_{\text{data}}(\mathbf{x}) = p_{\boldsymbol{\theta}}(\mathbf{x}), \quad D^*(\mathbf{x}) = 0.5.
$$

GAN Optimality

Theorem

The following minimax game

$$\min_G \max_D \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D(\mathbf{G}(\mathbf{z}))) \right]$$

achieves its global optimum precisely when $p_{\text{data}}(\mathbf{x}) = p_{\boldsymbol{\theta}}(\mathbf{x})$, and $D^*(\mathbf{x}) = 0.5$.

Expectations

If the generator can express **any** function and the discriminator is **optimal** at every step, the generator **will converge** to the target distribution.

Reality

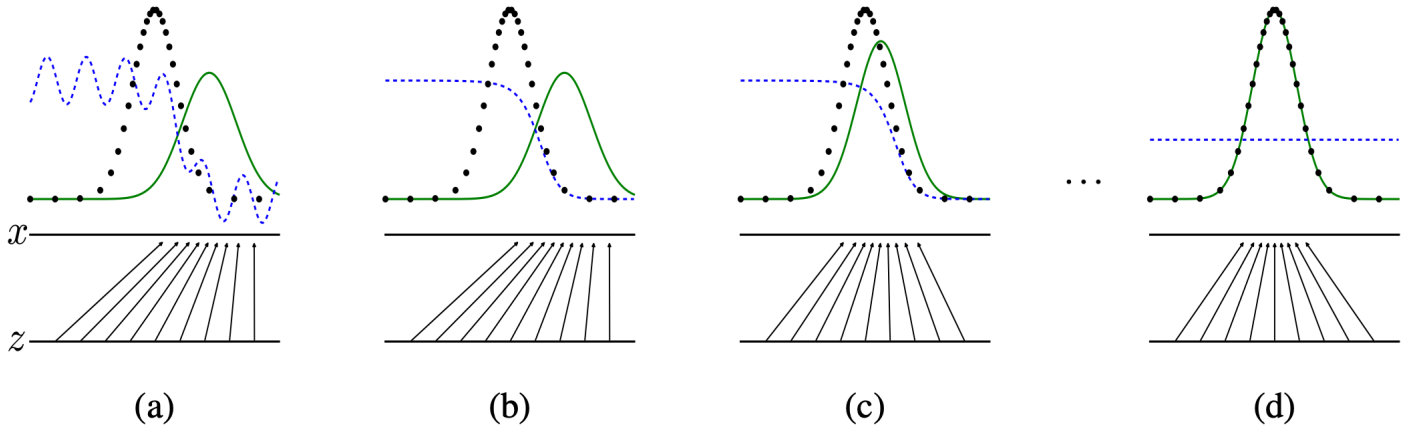
- Generator updates are performed in parameter space, and the discriminator is often imperfectly optimized.
- Generator and discriminator losses typically oscillate during GAN training.

GAN Training

Assume both generator and discriminator are parametric models: $D_{\boldsymbol{\phi}}(\mathbf{x})$ and $\mathbf{G}_{\boldsymbol{\theta}}(\mathbf{z})$.

Objective

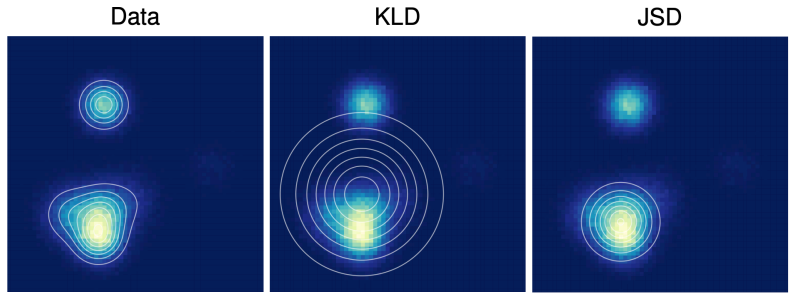
$$\min_{\boldsymbol{\theta}} \max_{\boldsymbol{\phi}} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D_{\boldsymbol{\phi}}(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D_{\boldsymbol{\phi}}(\mathbf{G}_{\boldsymbol{\theta}}(\mathbf{z})))\right]$$

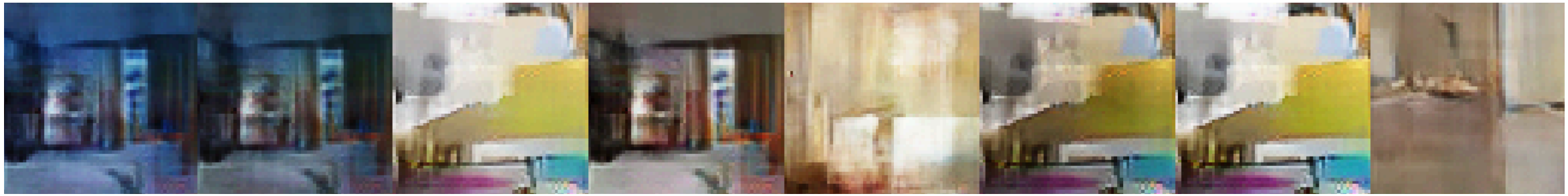


- $\mathbf{z} \sim p(\mathbf{z})$ is a latent variable.
- $p_{\boldsymbol{\theta}}(\mathbf{x}|\mathbf{z}) = \delta(\mathbf{x} - \mathbf{G}_{\boldsymbol{\theta}}(\mathbf{z}))$ serves as a deterministic decoder (like normalizing flows).
- There is no encoder present.

Mode Collapse

Mode collapse refers to the phenomenon where the generator in a GAN produces only one or a few different modes of the distribution.





Numerous methods have been proposed to tackle mode collapse: changing architectures, adding regularization terms, injecting noise.

Goodfellow I. J. et al. Generative Adversarial Networks, 2014
Metz L. et al. Unrolled Generative Adversarial Networks, 2016

Jensen-Shannon vs Kullback-Leibler Divergences

- $p_{\text{data}}(\mathbf{x})$ is a fixed mixture of two Gaussians.
- $p(\mathbf{x}|\mu, \sigma) = \mathcal{N}(\mu, \sigma^2)$.

Mode Covering vs. Mode Seeking

$$D_{\text{KL}}(\pi \,\|\, p) = \int \pi(\mathbf{x}) \log \frac{\pi(\mathbf{x})}{p(\mathbf{x})} d\mathbf{x}, \quad D_{\text{KL}}(p \| \pi) = \int p(\mathbf{x}) \log \frac{p(\mathbf{x})}{\pi(\mathbf{x})} d\mathbf{x}$$

$$D_{\text{JS}}(\pi \,\|\, p) = \frac{1}{2}\left[D_{\text{KL}}\left(\pi(\mathbf{x}) \,\|\, \frac{\pi(\mathbf{x}) + p(\mathbf{x})}{2}\right) + D_{\text{KL}}\left(p(\mathbf{x}) \,\|\, \frac{\pi(\mathbf{x}) + p(\mathbf{x})}{2}\right)\right]$$

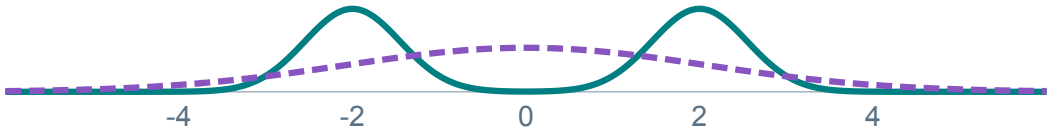
- **Forward KL** averages over data; **reverse KL** averages over model samples.
- **JSD** compares both distributions to their equally weighted mixture.

> Next: fit one Gaussian with the two KL directions.

Mode Covering vs. Mode Seeking

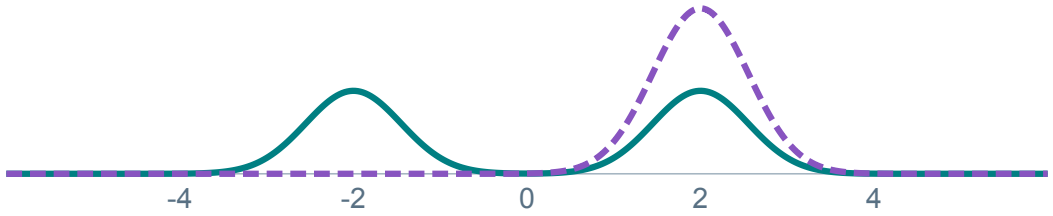
$$p_{\text{data}}(x) = \tfrac{1}{2}\mathcal{N}(x \mid -2, 0.55^2) + \tfrac{1}{2}\mathcal{N}(x \mid 2, 0.55^2), \qquad p_{\boldsymbol{\theta}}(x) = \mathcal{N}(x \mid \mu, \sigma^2).$$

Forward KL optimum



μ = 0.00, σ = 2.07
Forward KL 0.635 · Reverse KL 1.623

Reverse KL minimum (right mode)



μ = 2.00, σ = 0.55
Forward KL 12.445 · Reverse KL 0.693

Target: two modes Model: one Gaussian

Forward KL
Missing a target mode is expensive.

Reverse KL
Putting model mass in the low-density gap is expensive.

> This difference comes from restricting the model to one Gaussian. If it could match the target exactly, both KLs would be zero.

From VQ-VAE to VQGAN

Same **encoder → quantization → decoder**; a new reconstruction objective.



Same spatial downsampling $f = 8$ and codebook size $K = 8192$.

Perceptual Loss

A **fixed pretrained network** F extracts features:

$$\mathcal{L}_{\text{perc}} = \|F(\mathbf{x}) - F(\hat{\mathbf{x}})\|_2^2.$$

Adversarial Loss

Train discriminator D to separate real and reconstructed **image patches**.

Train the autoencoder to **fool** D.

Esser P., Rombach R., Ommer B. Taming Transformers for High-Resolution Image Synthesis, 2021. Figure 12.

Generative Modeling in Latent Space

First learn a compact representation; then learn to **generate its latent codes**.

Discrete: VQGAN

Quantization gives a map of code indices.

$$\mathbf{c} \in \{1, \dots, K\}^{W \times H}$$

Autoregressive prior over the code map.

$$\mathbf{c} \sim p_{\psi}(\mathbf{c}) \longrightarrow \mathbf{z}_q(\mathbf{c}) \longrightarrow \hat{\mathbf{x}}$$

Sample codes → lookup → decode.

Continuous: KL-Autoencoder

Perceptual + adversarial losses; weak KL.

$$\mathbf{z} \in \mathbb{R}^{W \times H \times L}$$

Diffusion prior in the learned latent space.

$$\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \longrightarrow \mathbf{z} \sim p_{\psi}(\mathbf{z}) \longrightarrow \hat{\mathbf{x}}$$

Denoise in latent space → decode.

> Both recipes freeze the autoencoder before learning a prior that models its **aggregated posterior**.

Esser P. et al. Taming Transformers for High-Resolution Image Synthesis, 2021.
Rombach R. et al. High-Resolution Image Synthesis with Latent Diffusion Models, 2022.

Outline

01 Generative Adversarial Networks (GAN)

02 Wasserstein Distance

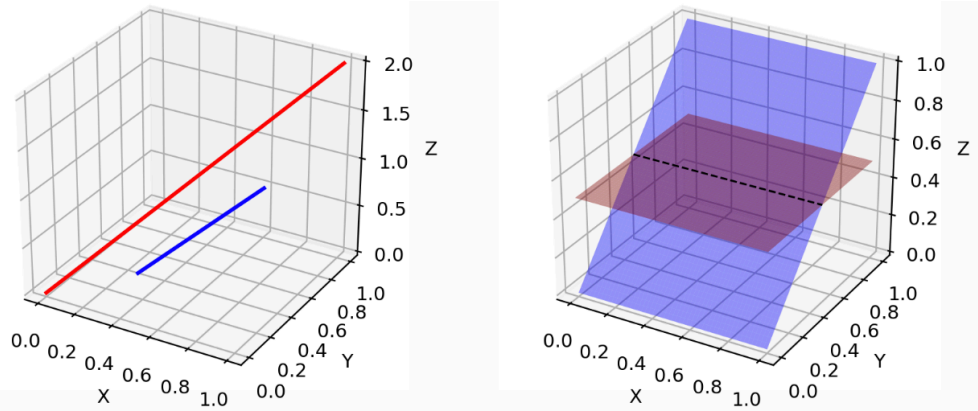
03 Wasserstein GAN (WGAN)

04 Evaluation of Likelihood-Free Models

Frechet Inception Distance (FID) · Precision-Recall · CLIP Score · Human Eval

Theoretical Results

- The dimensionality of $\mathbf{z}$ is less than that of $\mathbf{x}$, so $p_{\boldsymbol{\theta}}(\mathbf{x})$ with $\mathbf{x} = \mathbf{G}_{\boldsymbol{\theta}}(\mathbf{z})$ lives on a low-dimensional manifold.
- The true data distribution $p_{\text{data}}(\mathbf{x})$ is also supported on a low-dimensional manifold.



- If $p_{\text{data}}(\mathbf{x})$ and $p_{\boldsymbol{\theta}}(\mathbf{x})$ are disjoint, a smooth optimal discriminator can exist!

For such low-dimensional, disjoint manifolds:

$$D_{\text{KL}}(p_{\text{data}} \,\|\, p_{\boldsymbol{\theta}}) = D_{\text{KL}}(p_{\boldsymbol{\theta}} \,\|\, p_{\text{data}}) = \infty, \quad D_{\text{JS}}(p_{\text{data}} \,\|\, p_{\boldsymbol{\theta}}) = \log 2$$

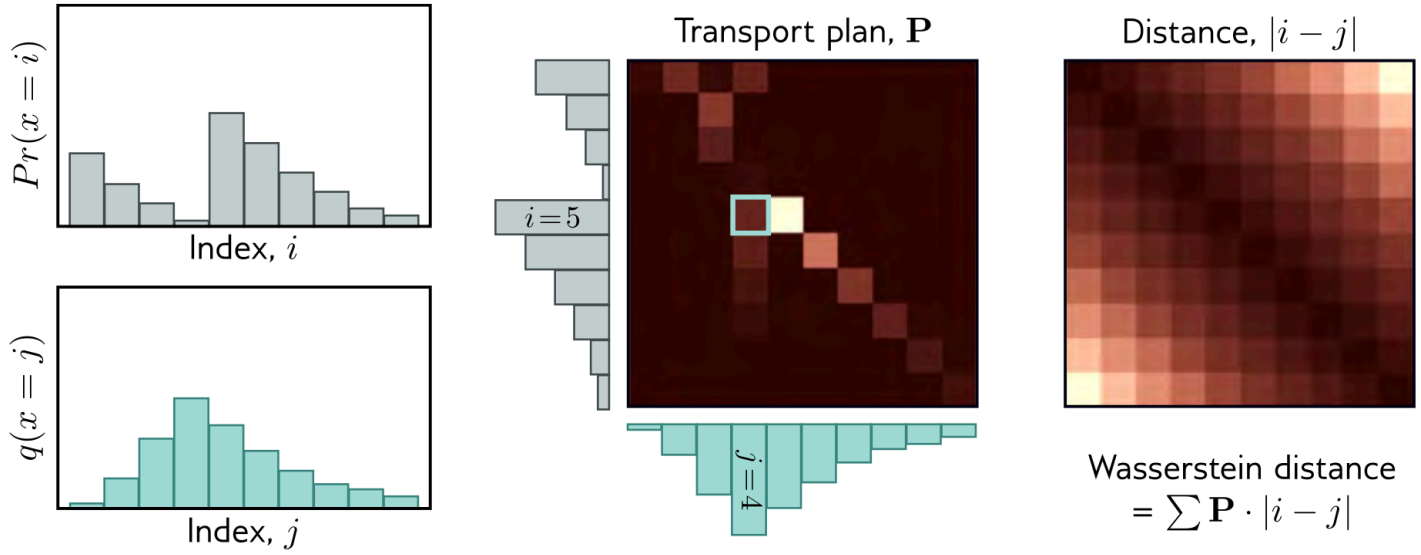
Weng L. From GAN to WGAN, 2019
Arjovsky M., Bottou L. Towards Principled Methods for Training Generative Adversarial Networks, 2017

Wasserstein Distance (Discrete)

Also known as the **Earth Mover's Distance**.

Optimal Transport Formulation

The minimum cost of moving and transforming a pile of "dirt" shaped like one probability distribution to match another.



Wasserstein Distance (Continuous)

$$W(\pi\|p) = \inf_{\gamma\in\Gamma(\pi,p)} \mathbb{E}_{(\mathbf{x}_1,\mathbf{x}_2)\sim\gamma}\|\mathbf{x}_1 - \mathbf{x}_2\|$$

$$= \inf_{\gamma\in\Gamma(\pi,p)} \int \|\mathbf{x}_1 - \mathbf{x}_2\|\,\gamma(\mathbf{x}_1,\mathbf{x}_2)d\mathbf{x}_1 d\mathbf{x}_2$$

- $\gamma(\mathbf{x}_1,\mathbf{x}_2)$ is the transport plan: the amount of "dirt" assigned from $\mathbf{x}_1$ to $\mathbf{x}_2$.

$$\int \gamma(\mathbf{x}_1,\mathbf{x}_2)d\mathbf{x}_1 = p(\mathbf{x}_2); \quad \int \gamma(\mathbf{x}_1,\mathbf{x}_2)d\mathbf{x}_2 = \pi(\mathbf{x}_1).$$

- $\Gamma(\pi,p)$ denotes the set of all joint distributions $\gamma(\mathbf{x}_1,\mathbf{x}_2)$ with marginals π and p.
- $\gamma(\mathbf{x}_1,\mathbf{x}_2)$ is the mass, $\|\mathbf{x}_1 - \mathbf{x}_2\|$ is the distance.

Wasserstein Metric

$$W_s(\pi,p) = \inf_{\gamma\in\Gamma(\pi,p)} \left(\mathbb{E}_{(\mathbf{x}_1,\mathbf{x}_2)\sim\gamma}\|\mathbf{x}_1 - \mathbf{x}_2\|^s\right)^{1/s}$$

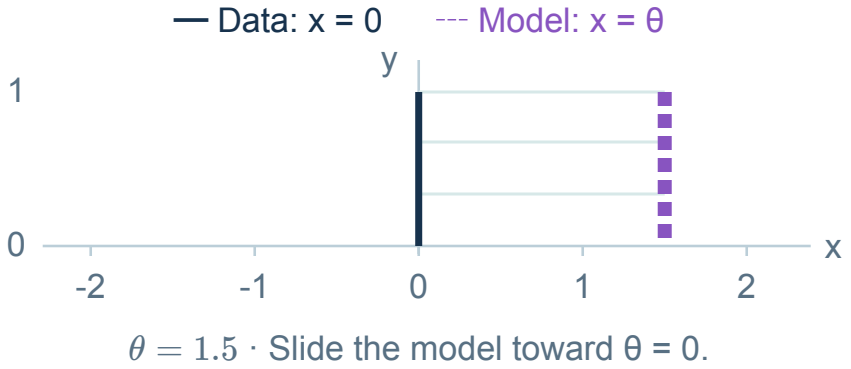
In our setting, $W(\pi\|p) = W_1(\pi,p)$, which is the transport cost formulation.

In Lecture 12, minibatch optimal transport pairs noise and data samples for flow matching.

Wasserstein Distance vs KL vs JSD

Consider two-dimensional distributions:

$$p_{\text{data}}(x, y) = (0, U[0, 1]), \qquad p_{\boldsymbol{\theta}}(x, y) = (\theta, U[0, 1])$$



$\theta = 0$: Both distributions are identical.

$$D_{\text{KL}}(p_{\text{data}} \| p_{\boldsymbol{\theta}}) = D_{\text{KL}}(p_{\boldsymbol{\theta}} \| p_{\text{data}}) = D_{\text{JS}}(p_{\boldsymbol{\theta}} \| p_{\text{data}}) = W(p_{\text{data}} \| p_{\boldsymbol{\theta}}) = 0$$

$\theta \neq 0$:

$$D_{\text{KL}}(p_{\text{data}} \| p_{\boldsymbol{\theta}}) = \int_{U[0,1]} 1 \log \frac{1}{0} \, dy = \infty = D_{\text{KL}}(p_{\boldsymbol{\theta}} \| p_{\text{data}})$$

$$D_{\text{JS}}(p_{\text{data}} \| p_{\boldsymbol{\theta}}) = \frac{1}{2} \left(\int_{U[0,1]} 1 \log \frac{1}{1/2} dy + \int_{U[0,1]} 1 \log \frac{1}{1/2} dy \right) = \log 2$$

$$W(p_{\text{data}} \| p_{\boldsymbol{\theta}}) = |\theta|$$

Weng L. From GAN to WGAN, 2019
Arjovsky M., Chintala S., Bottou L. Wasserstein GAN, 2017

Why Wasserstein Distance?

On a compact data space $\mathcal{X}$:

Dependence on Generator Parameters

For a feedforward generator $\mathbf{G}_{\boldsymbol{\theta}}$ with Lipschitz activations and $\mathbb{E}_{p(\mathbf{z})}\|\mathbf{z}\| < \infty$, $W(p_{\text{data}}\|p_{\boldsymbol{\theta}})$ is continuous in $\boldsymbol{\theta}$ and differentiable almost everywhere.

Convergence of Distributions

For a fixed distribution π and a sequence p_t on $\mathcal{X}$, as $t \to \infty$:

$$D_{\text{KL}}(\pi\|p_t) \to 0 \quad \Longrightarrow \quad D_{\text{JS}}(\pi\|p_t) \to 0 \quad \Longrightarrow \quad W(\pi\|p_t) \to 0$$

The first implication also holds for $D_{\text{KL}}(p_t\|\pi) \to 0$; the converses need not hold.

> **Wasserstein can capture distributions getting closer even when their supports do not overlap.**

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Wasserstein GAN

Wasserstein Distance

$$W(\pi \| p) = \inf_{\gamma \in \Gamma(\pi, p)} \mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \|\mathbf{x}_1 - \mathbf{x}_2\|$$

$$= \inf_{\gamma \in \Gamma(\pi, p)} \int \|\mathbf{x}_1 - \mathbf{x}_2\| \gamma(\mathbf{x}_1, \mathbf{x}_2) \, d\mathbf{x}_1 \, d\mathbf{x}_2$$

The infimum over all possible $\gamma \in \Gamma(\pi, p)$ is computationally intractable.

Theorem (Kantorovich-Rubinstein Duality)

$$W(\pi \| p) = \frac{1}{K} \max_{\|f\|_L \leq K} \Big[\mathbb{E}_{\pi(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p(\mathbf{x})} f(\mathbf{x}) \Big]$$

where $f : \mathbb{R}^m \to \mathbb{R}$ is K-Lipschitz ($\|f\|_L \leq K$):

$$|f(\mathbf{x}_1) - f(\mathbf{x}_2)| \leq K \|\mathbf{x}_1 - \mathbf{x}_2\|, \quad \forall \, \mathbf{x}_1, \mathbf{x}_2 \in \mathcal{X}.$$

We can thus estimate $W(\pi \| p)$ using only samples and a function f.

Wasserstein GAN

Theorem (Kantorovich-Rubinstein Duality)

$$W(p_{\text{data}} \| p_{\boldsymbol{\theta}}) = \frac{1}{K} \max_{\|f\|_L \leq K} \Big[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} f(\mathbf{x}) \Big]$$

- We must ensure that f is K-Lipschitz continuous.
- Let $f_{\boldsymbol{\phi}}(\mathbf{x})$ be a feedforward neural network parameterized by $\boldsymbol{\phi}$.
- If the weights $\boldsymbol{\phi}$ are restricted to a compact set $\boldsymbol{\Phi}$, then $f_{\boldsymbol{\phi}}$ is K-Lipschitz.
- Clamp weights within the box $\boldsymbol{\Phi} = [-c, c]^d$ (e.g. $c = 0.01$) after each update.

$$\begin{aligned} K \cdot W(p_{\text{data}} \| p_{\boldsymbol{\theta}}) &= \max_{\|f\|_L \leq K} \Big[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} f(\mathbf{x}) \Big] \\ &\geq \max_{\boldsymbol{\phi} \in \boldsymbol{\Phi}} \Big[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f_{\boldsymbol{\phi}}(\mathbf{x}) - \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} f_{\boldsymbol{\phi}}(\mathbf{x}) \Big] \end{aligned}$$

Wasserstein GAN

Standard GAN Objective

$$\min_{\boldsymbol{\theta}} \max_{\boldsymbol{\phi}} \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D_{\boldsymbol{\phi}}(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D_{\boldsymbol{\phi}}(\mathbf{G}_{\boldsymbol{\theta}}(\mathbf{z})))$$

WGAN Objective

$$\min_{\boldsymbol{\theta}} W(p_{\text{data}} \| p_{\boldsymbol{\theta}}) \approx \min_{\boldsymbol{\theta}} \max_{\boldsymbol{\phi} \in \boldsymbol{\Phi}} \left[\mathbb{E}_{p_{\text{data}}(\mathbf{x})} f_{\boldsymbol{\phi}}(\mathbf{x}) - \mathbb{E}_{p(\mathbf{z})} f_{\boldsymbol{\phi}}(\mathbf{G}_{\boldsymbol{\theta}}(\mathbf{z})) \right]$$

- The discriminator D is replaced by function f: in WGAN, it is known as the **critic**, which is *not* a classifier.
- *"Weight clipping is a clearly terrible way to enforce a Lipschitz constraint."*
  - If c is large, optimizing the critic is hard; if c is small, gradients may vanish.

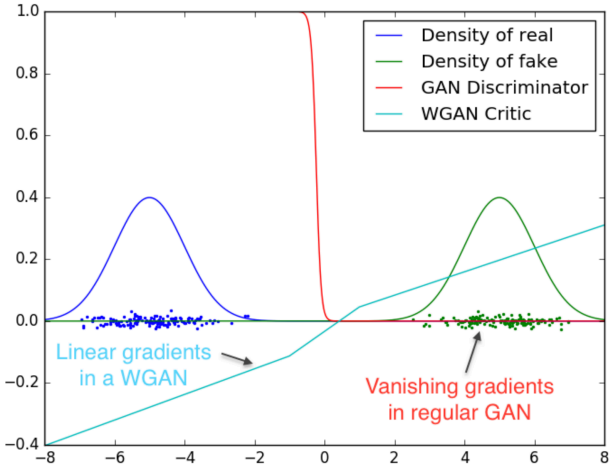
Gradient penalty encourages the critic's input-gradient norm to stay close to 1 on interpolations between real and generated samples.

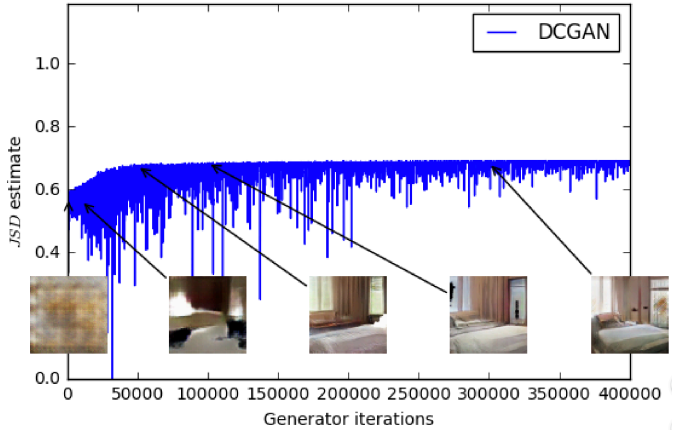
Spectral normalization controls layer-wise Lipschitz constants by dividing weight matrices by their largest singular values.

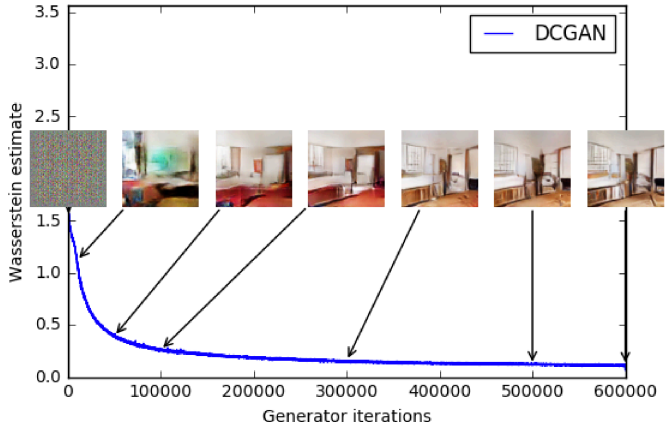
Arjovsky M., Chintala S., Bottou L. Wasserstein GAN, 2017
Gulrajani I. et al. Improved Training of Wasserstein GANs, 2017
Miyato T. et al. Spectral Normalization for Generative Adversarial Networks, 2018

Wasserstein GAN

- WGAN provides nonzero gradients even if distributions' supports are disjoint.
- $D_{\mathrm{JS}}(p_{\mathrm{data}} \| p_{\boldsymbol{\theta}})$ is poorly correlated with sample quality and remains near its maximum value $\log 2 \approx 0.69$.
- $W(p_{\mathrm{data}} \| p_{\boldsymbol{\theta}})$ is tightly correlated with quality.







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What Makes a Good Generator?

Likelihood alone does not tell us whether samples are useful for the target task.

Fidelity

Does each sample look plausible?



Sharp, but anatomically wrong.

Coverage & Diversity

Does the set cover the data?



The same output on every run.

Condition Alignment

Does it follow the condition?



Prompt: “Three red apples.”
Result: only two.

Illustrative failure cases created for this lecture.

Theis L. et al. A Note on the Evaluation of Generative Models, 2016 · Kynkäänniemi T. et al. Improved Precision and Recall Metric for Assessing Generative Models, 2019
Lin Z. et al. Evaluating Text-to-Visual Generation with Image-to-Text Generation, 2024

Outline

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Wasserstein GAN (Optimal Transport)

$$\begin{aligned} W(\pi \| p) &= \inf_{\gamma \in \Gamma(\pi, p)} \mathbb{E}_{(\mathbf{x}_1, \mathbf{x}_2) \sim \gamma} \| \mathbf{x}_1 - \mathbf{x}_2 \| \\ &= \inf_{\gamma \in \Gamma(\pi, p)} \int \| \mathbf{x}_1 - \mathbf{x}_2 \| \gamma(\mathbf{x}_1, \mathbf{x}_2) \, d\mathbf{x}_1 \, d\mathbf{x}_2 \end{aligned}$$

Theorem

If $\pi(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_\pi, \boldsymbol{\Sigma}_\pi)$, $p(\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_p, \boldsymbol{\Sigma}_p)$, then

$$W_2^2(\pi \| p) = \| \boldsymbol{\mu}_\pi - \boldsymbol{\mu}_p \|^2 + \operatorname{tr} \left[\boldsymbol{\Sigma}_\pi + \boldsymbol{\Sigma}_p - 2 \left(\boldsymbol{\Sigma}_\pi^{1/2} \boldsymbol{\Sigma}_p \boldsymbol{\Sigma}_\pi^{1/2} \right)^{1/2} \right]$$

Frechet Inception Distance

$$\mathrm{FID}(p_{\mathrm{data}}, p_{\boldsymbol{\theta}}) = W_2^2(p_{\mathrm{data}} \| p_{\boldsymbol{\theta}})$$

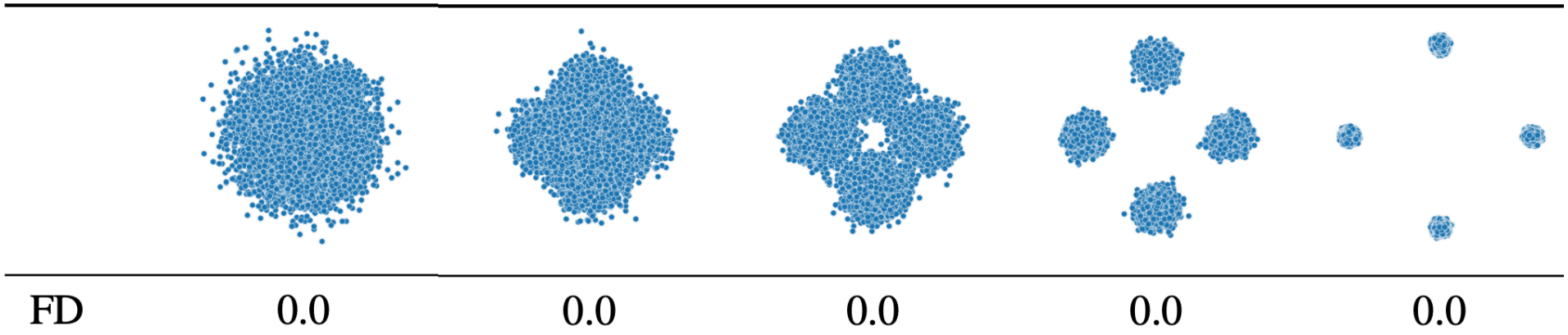
Heusel M. et al. GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium, 2017

Frechet Inception Distance (FID)

$$\mathrm{FID}(p_{\text{data}}, p_{\boldsymbol{\theta}}) = \|\boldsymbol{\mu}_{\text{data}} - \boldsymbol{\mu}_{\boldsymbol{\theta}}\|^2 + \mathrm{tr}\left[\boldsymbol{\Sigma}_{\text{data}} + \boldsymbol{\Sigma}_{\boldsymbol{\theta}} - 2\left(\boldsymbol{\Sigma}_{\text{data}}^{1/2}\boldsymbol{\Sigma}_{\boldsymbol{\theta}}\boldsymbol{\Sigma}_{\text{data}}^{1/2}\right)^{1/2}\right]$$

- FID uses Inception image embeddings $\mathbf{z} = \mathbf{f}(\mathbf{x})$.
- $\boldsymbol{\mu}_{\text{data}}$, $\boldsymbol{\Sigma}_{\text{data}}$ and $\boldsymbol{\mu}_{\boldsymbol{\theta}}$, $\boldsymbol{\Sigma}_{\boldsymbol{\theta}}$ are statistics of latent representations for samples from $p_{\text{data}}(\mathbf{x})$ and $p_{\boldsymbol{\theta}}(\mathbf{x})$.
- **FD-DINOv2** uses the same formula with DINOv2 features; the Gaussian approximation remains.

$\mathrm{FID}(p(\mathbf{x}), \mathcal{N}(0, \mathbf{I}))$



Jayasumana S. et al. Rethinking FID: Towards a Better Evaluation Metric for Image Generation, 2024
Stein G. et al. Exposing Flaws of Generative Model Evaluation Metrics and Their Unfair Treatment of Diffusion Models, 2023

Frechet Inception Distance (FID)

$$\mathrm{FID}(p_{\text{data}}, p_{\boldsymbol{\theta}}) = \|\boldsymbol{\mu}_{\text{data}} - \boldsymbol{\mu}_{\boldsymbol{\theta}}\|^2 + \mathrm{tr}\left[\boldsymbol{\Sigma}_{\text{data}} + \boldsymbol{\Sigma}_{\boldsymbol{\theta}} - 2\left(\boldsymbol{\Sigma}_{\text{data}}^{1/2}\boldsymbol{\Sigma}_{\boldsymbol{\theta}}\boldsymbol{\Sigma}_{\text{data}}^{1/2}\right)^{1/2}\right]$$

Drawbacks

- Depends on the pretrained classification network.
- Uses the normality assumption.
- May not correlate with human evaluation.

Finite-sample FID is biased, and its value depends on the sample size.

Model	Model-A	Model-B
FID	21.40	18.42
FID_∞	20.16	17.19
Human rater preference	92.5%	6.9%

Jayasumana S. et al. Rethinking FID: Towards a Better Evaluation Metric for Image Generation, 2024

Maximum Mean Discrepancy (MMD)

The earlier figure shows that matching means and covariances can miss changes in shape. For embedding distributions P, Q, compare means of richer nonlinear features ψ:

$$\boldsymbol{\mu}_P = \mathbb{E}_{\mathbf{z}\sim P}\psi(\mathbf{z}), \qquad \boldsymbol{\mu}_Q = \mathbb{E}_{\mathbf{w}\sim Q}\psi(\mathbf{w}).$$

Kernel Trick

$$k(\mathbf{a}, \mathbf{u}) = \langle \psi(\mathbf{a}), \psi(\mathbf{u}) \rangle.$$

Compute inner products directly, without explicitly constructing ψ.

Distance Between Mean Features

Let $\mathbf{z}, \mathbf{z}' \sim P$ and $\mathbf{w}, \mathbf{w}' \sim Q$ be mutually independent.

$$\begin{aligned}
\mathrm{MMD}_k^2(P, Q) &= \|\boldsymbol{\mu}_P - \boldsymbol{\mu}_Q\|^2 \\
&= \langle \boldsymbol{\mu}_P, \boldsymbol{\mu}_P \rangle + \langle \boldsymbol{\mu}_Q, \boldsymbol{\mu}_Q \rangle - 2\langle \boldsymbol{\mu}_P, \boldsymbol{\mu}_Q \rangle \\
&= \underbrace{\mathbb{E}k(\mathbf{z}, \mathbf{z}')}_{\text{within } P} + \underbrace{\mathbb{E}k(\mathbf{w}, \mathbf{w}')}_{\text{within } Q} - 2\,\underbrace{\mathbb{E}k(\mathbf{z}, \mathbf{w})}_{\text{between } P,Q}.
\end{aligned}$$

CMMD: Comparing CLIP Image Embeddings

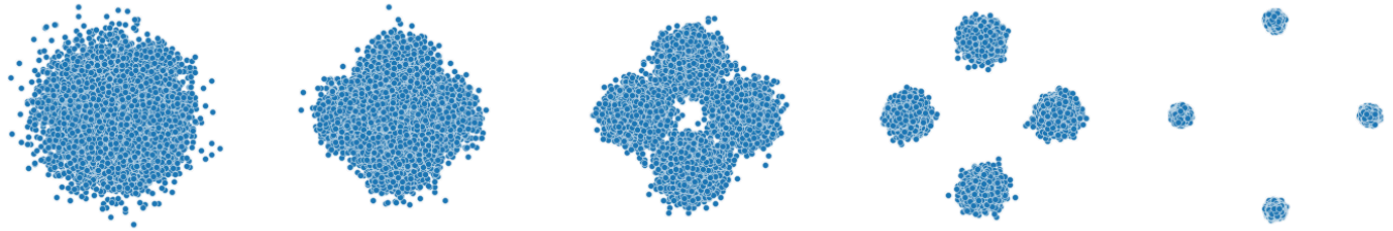
CMMD = squared MMD on **CLIP image embeddings**.

$$\mathrm{MMD}_k^2(P, Q) = \mathbb{E}k(\mathbf{z}, \mathbf{z}') + \mathbb{E}k(\mathbf{w}, \mathbf{w}') - 2\mathbb{E}k(\mathbf{z}, \mathbf{w}).$$

$$k(\mathbf{z}, \mathbf{w}) = \exp\left(-\frac{\|\mathbf{z} - \mathbf{w}\|^2}{2\sigma^2}\right), \qquad \sigma > 0.$$

One Gaussian kernel already corresponds to infinitely many nonlinear features in ψ.

$$\mathrm{MMD}_k(P, Q) = 0 \quad \Longleftrightarrow \quad P = Q.$$



FD	0.0	0.0	0.0	0.0	0.0
MMD	0.0	5.794	17.21	78.88	202.8

Synthetic example: same means and covariances, different shapes.

Outline

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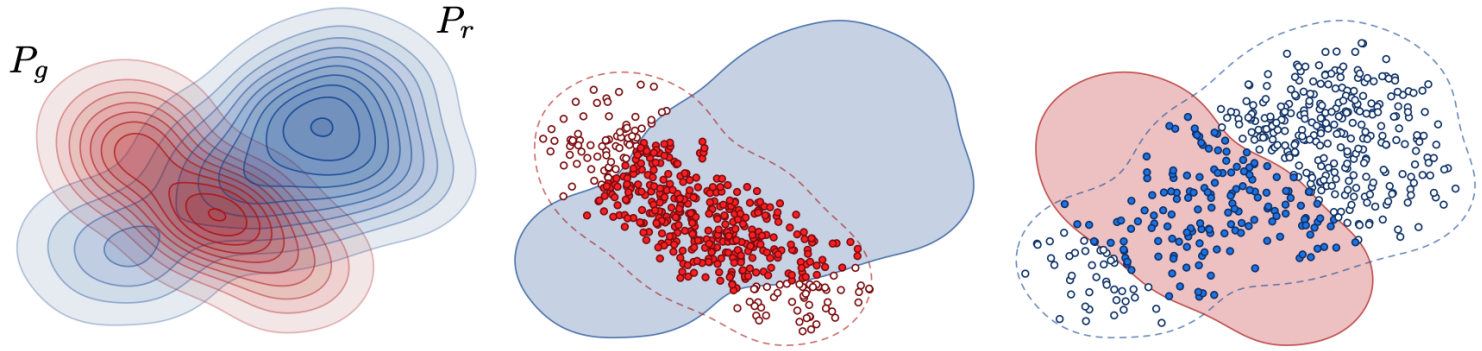
04 Evaluation of Likelihood-Free Models

Frechet Inception Distance (FID) · **Precision-Recall** · CLIP Score · Human Eval

Precision-Recall

Desirable Properties for Samples

- **Sharpness:** generated samples should possess high visual quality.
- **Diversity:** their variation should match that in the training data.



(a) Example distributions (b) Precision (c) Recall

- **Precision** denotes the fraction of generated images that look realistic.
- **Recall** measures how well the generator covers the training data manifold.

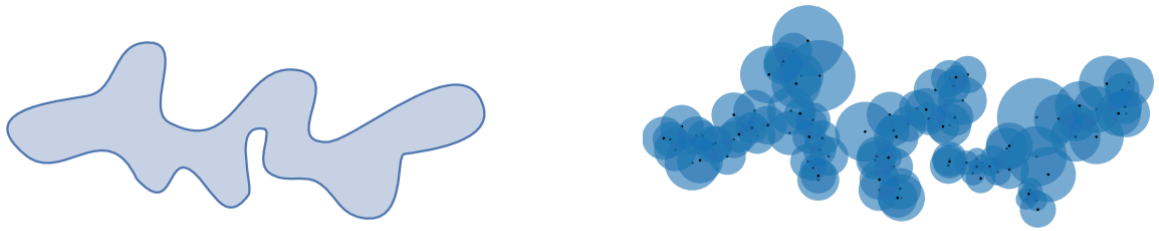
Precision-Recall

- $\mathcal{S}_{\text{data}} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\text{data}}(\mathbf{x})$ — real samples;
- $\mathcal{S}_{\boldsymbol{\theta}} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\boldsymbol{\theta}}(\mathbf{x})$ — generated samples.

Define a binary function:

$$\mathbb{I}(\mathbf{x}, \mathcal{S}) = \begin{cases} 1, & \text{if } \exists\, \mathbf{x}' \in \mathcal{S} : \|\mathbf{x} - \mathbf{x}'\|_2 \leq \|\mathbf{x}' - \text{NN}_k(\mathbf{x}', \mathcal{S})\|_2; \\ 0, & \text{otherwise.} \end{cases}$$

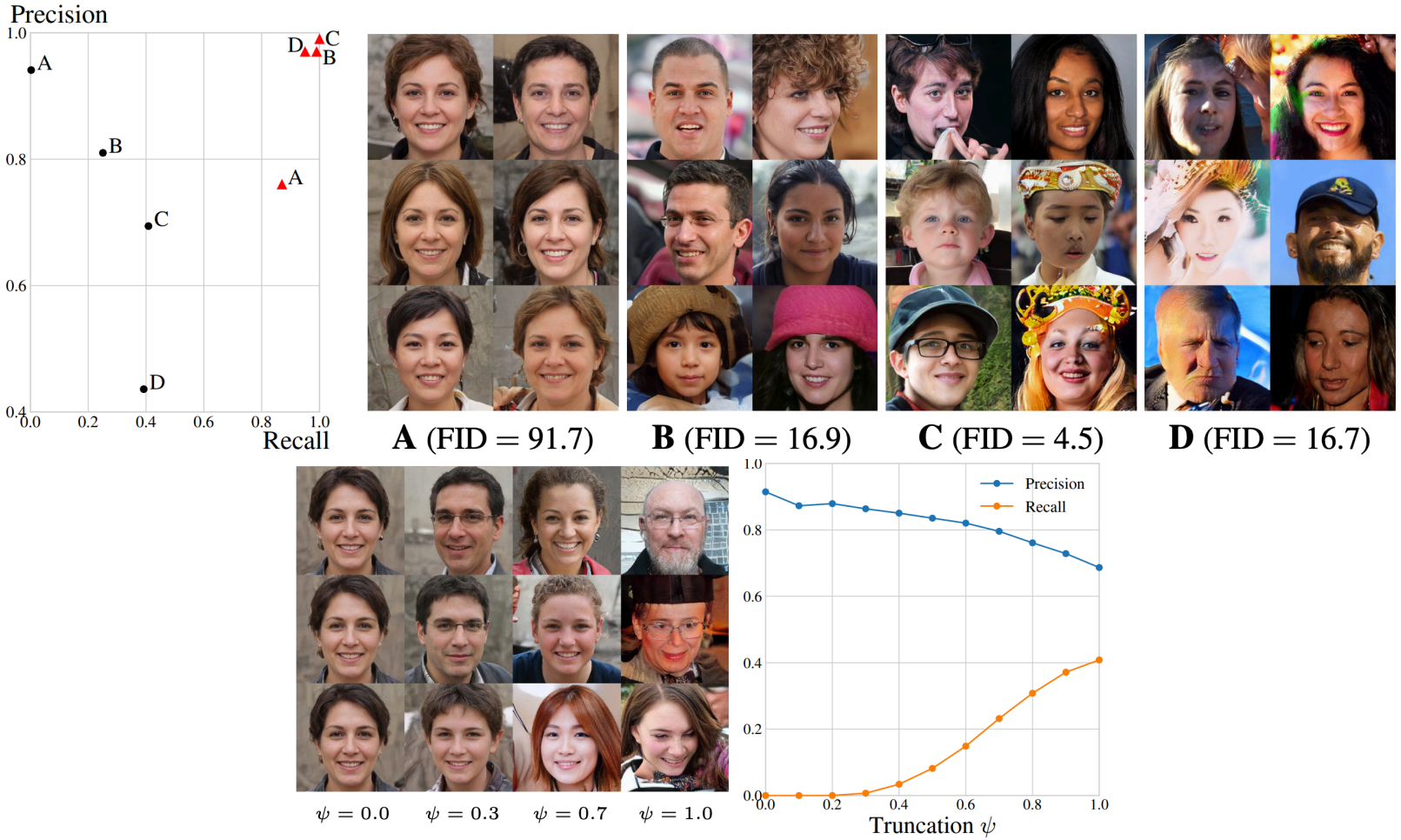
$$\text{Pr}(\mathcal{S}_{\text{data}}, \mathcal{S}_{\boldsymbol{\theta}}) = \frac{1}{n} \sum_{\mathbf{x} \in \mathcal{S}_{\boldsymbol{\theta}}} \mathbb{I}(\mathbf{x}, \mathcal{S}_{\text{data}}); \quad \text{Rec}(\mathcal{S}_{\text{data}}, \mathcal{S}_{\boldsymbol{\theta}}) = \frac{1}{n} \sum_{\mathbf{x} \in \mathcal{S}_{\text{data}}} \mathbb{I}(\mathbf{x}, \mathcal{S}_{\boldsymbol{\theta}}).$$



(a) True manifold (b) Approx. manifold

Embed the samples using a pretrained network (as in FID).

Precision-Recall



Here, stronger truncation (smaller ψ) favors precision over recall.

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CLIP Similarity and Human Preferences

Score an image $\mathbf{x}$ for a text prompt t.

CLIP: Learn from Image–Caption Pairs

Map images and text into a shared embedding space:

$$s_{\text{CLIP}}(\mathbf{x}, t) = \cos(\mathbf{f}_{\text{img}}(\mathbf{x}), \mathbf{f}_{\text{text}}(t)) \,.$$

PickScore: Learn from Human Choices

Fine-tune CLIP on pairs: for the same prompt, a human prefers $\mathbf{x}^+$ to $\mathbf{x}^-$.

$$\mathbb{P}_{\boldsymbol{\theta}}(\mathbf{x}^+ \succ \mathbf{x}^- \mid t) = \frac{\exp r_{\boldsymbol{\theta}}(\mathbf{x}^+, t)}{\exp r_{\boldsymbol{\theta}}(\mathbf{x}^+, t) + \exp r_{\boldsymbol{\theta}}(\mathbf{x}^-, t)}.$$

Train the score $r_{\boldsymbol{\theta}}$ to assign higher probability to the human choice.

Other preference models: **ImageReward**, **HPSv2**.

A learned preference score can guide sample selection and model training — but optimizing the score can exploit its flaws.

Radford A. et al. Learning Transferable Visual Models From Natural Language Supervision, 2021
Kirstain Y. et al. Pick-a-Pic: An Open Dataset of User Preferences for Text-to-Image Generation, 2023

Does the Image Follow the Prompt?

Prompt t: **“A red cube above a blue sphere.”**



Same objects and colors; opposite spatial relation. CLIP similarity can miss this.

VQAScore: Ask a Visual Question

$q(t)$: “Does this image show a red cube above a blue sphere?”

$$s_{\mathrm{VQA}}(\mathbf{x}, t) = p_{\mathrm{VQA}}(\mathrm{Yes} \mid \mathbf{x}, q(t)).$$

GenEval / DPG-Bench: test objects, counts, attributes, and relations across prompts.

Lin Z. et al. Evaluating Text-to-Visual Generation with Image-to-Text Generation, 2024
Ghosh D. et al. GenEval: An Object-Focused Framework for Evaluating Text-to-Image Alignment, 2023
Hu X. et al. ELLA: Equip Diffusion Models with LLM for Enhanced Semantic Alignment, 2024 (DPG-Bench)

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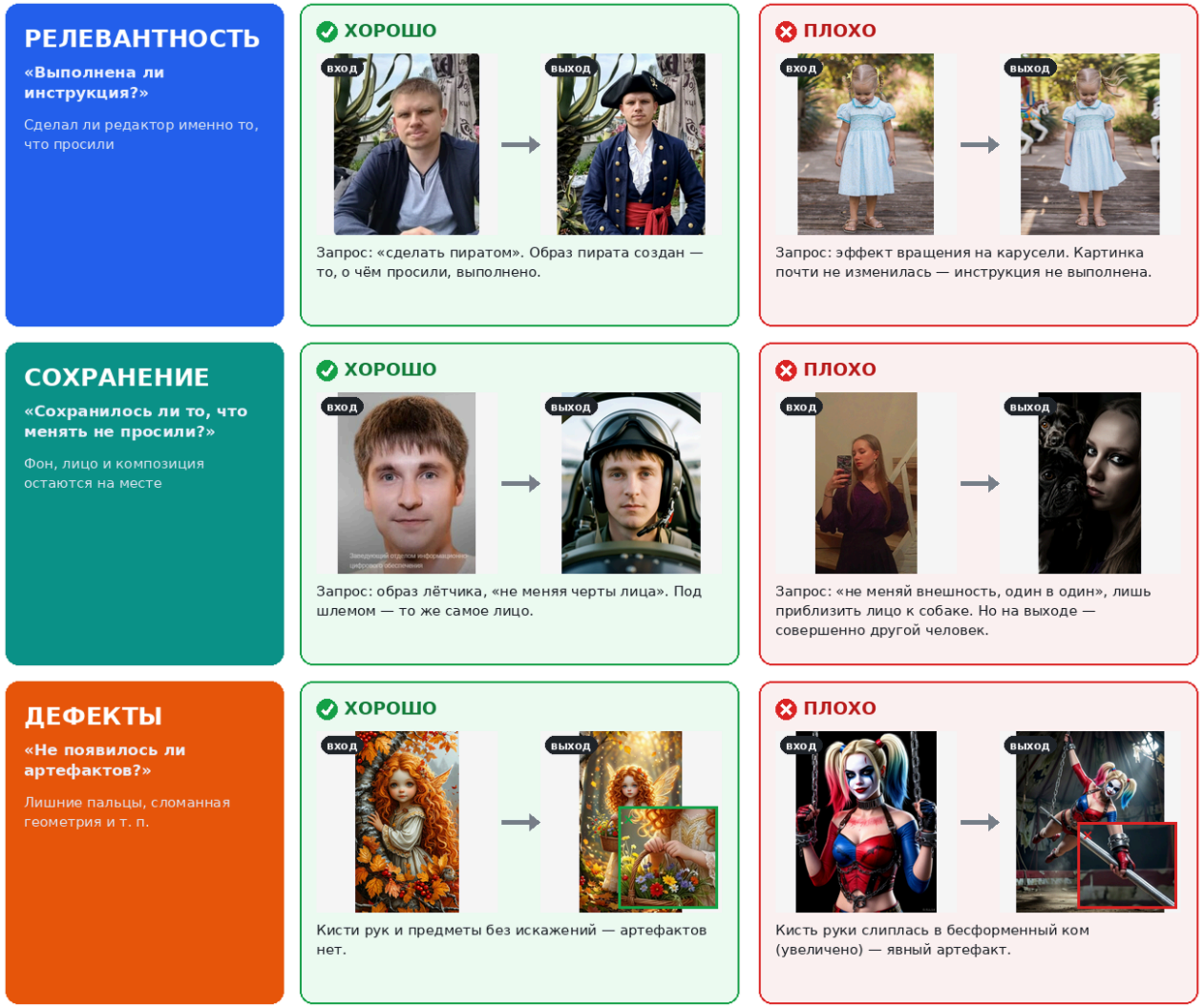
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Human Evaluation

- No automated metric is perfect.
- The best way to evaluate generative models is by human assessment.
- It’s important to assess various properties.



Summary

- With an optimal discriminator, the GAN minimax objective minimizes Jensen-Shannon divergence; alternating training can suffer from mode collapse.
- Perceptual and adversarial losses train autoencoders; learned priors model their latents.
- WGAN uses Kantorovich-Rubinstein duality to estimate a distance informative even for disjoint supports; weight clipping constrains the critic.
- FID compares Gaussian approximations; CMMD compares distributions of CLIP features using a kernel.
- Precision-recall measures the balance between sample quality and diversity.
- Image–text similarity, human preference, and compositional correctness require different evaluation signals.